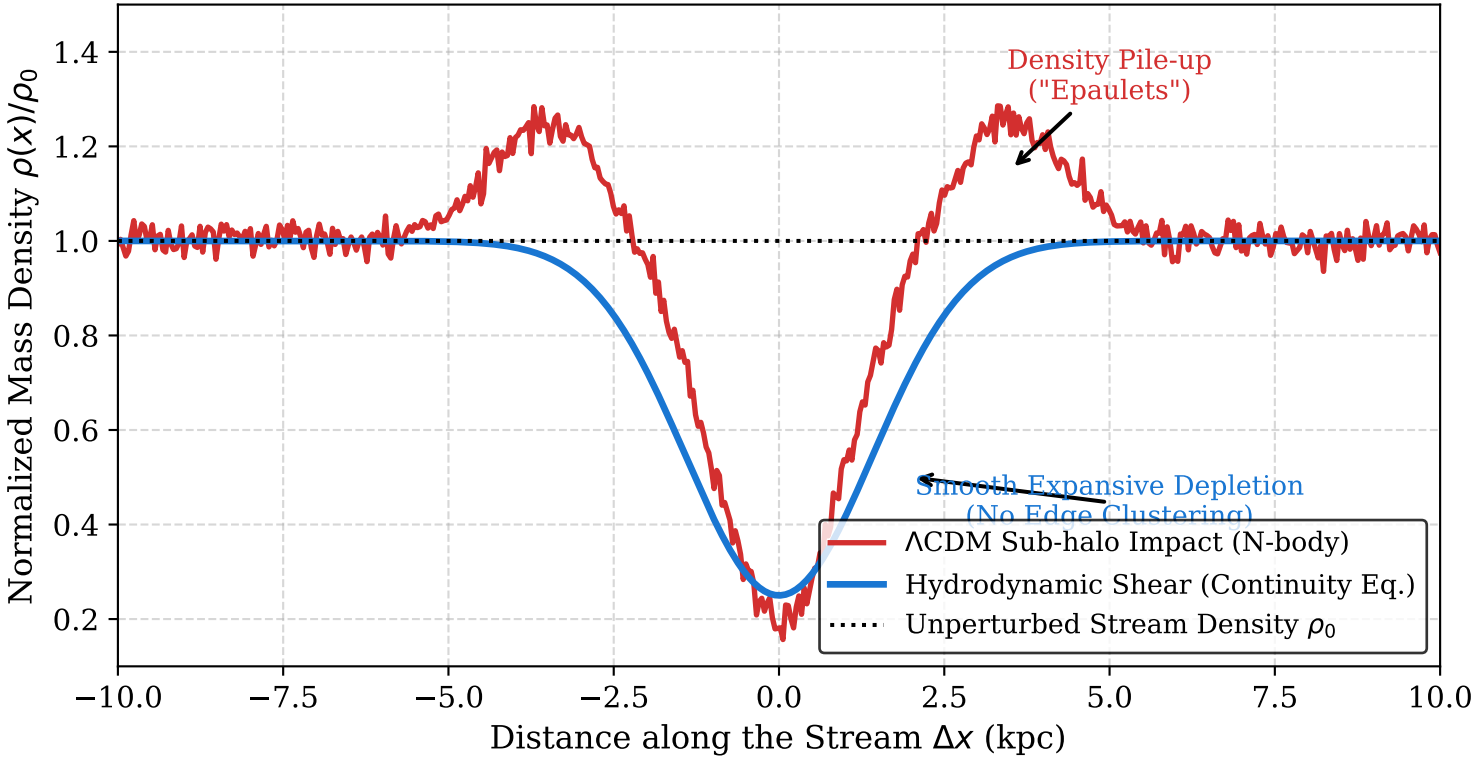
Morphological Density Profiles: N-body vs. Hydrodynamic Fluid
Density Pile-up
("Epaulets")
Smooth Expansive Depletion
(No Edge Clustering)
ΛCDM Sub-halo Impact (N-body)
Hydrodynamic Shear (Continuity Eq.)
Unperturbed Stream Density ρ_0
Normalized Mass Density $\rho(x)/\rho_0$
Distance along the Stream Δx (kpc)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0